Human vs AI Log Detective Activity

Scenario: The Campus Learning System Incident

Last night the OntarioTech Learning Portal experienced intermittent failures.
Students reported login problems, slow performance, and random logouts.
At 03:14 AM, the system restarted automatically.

The IT team extracted a portion of the system log file.
Your job is to determine what happened.

Possible Causes

- Software bug
- System overload
- Cyberattack
- Database failure

Sample Log File (learning_portal.log)

2026-02-18 02:58:01 INFO AuthService User login success user=jsmith ip=192.168.1.45

2026-02-18 02:59:10 INFO AuthService User login success user=ajohnson ip=192.168.1.52

2026-02-18 03:01:44 INFO CourseService Course page loaded course=INFR3610 user=jsmith

2026-02-18 03:05:13 WARN DatabaseService Query latency detected query_time=2.9s

2026-02-18 03:07:01 INFO AuthService User login success user=rpatel ip=192.168.1.88

2026-02-18 03:08:33 WARN DatabaseService Query latency detected query_time=3.2s

2026-02-18 03:09:22 INFO FileService File uploaded user=ajohnson file=assignment2.pdf

2026-02-18 03:10:05 ERROR DatabaseService Connection timeout db=CourseDB

2026-02-18 03:10:08 ERROR DatabaseService Connection timeout db=CourseDB

2026-02-18 03:10:12 ERROR DatabaseService Connection timeout db=CourseDB

2026-02-18 03:10:20 WARN SecurityService Multiple failed login attempts user=admin

2026-02-18 03:10:23 WARN SecurityService Failed login user=admin ip=45.33.22.91

2026-02-18 03:10:26 WARN SecurityService Failed login user=admin ip=45.33.22.91

2026-02-18 03:10:29 WARN SecurityService Failed login user=admin ip=45.33.22.91

2026-02-18 03:10:32 WARN SecurityService Failed login user=admin ip=45.33.22.91

2026-02-18 03:10:35 WARN SecurityService Failed login user=admin ip=45.33.22.91

2026-02-18 03:10:36 ALERT SecurityService Possible brute force attack detected ip=45.33.22.91

2026-02-18 03:10:40 ERROR AuthService Authentication service overloaded

2026-02-18 03:11:02 ERROR CourseService Failed to retrieve course data

2026-02-18 03:12:12 ERROR DatabaseService Connection pool exhausted

2026-02-18 03:13:50 CRITICAL SystemService System instability detected

2026-02-18 03:14:00 SYSTEM Kernel Restart initiated

2026-02-18 03:14:01 INFO SystemService System reboot complete

Student Questions

1. What anomalies do you observe in the logs?
2. Which IP address appears suspicious?
3. When did the incident begin?
4. What do you think caused the system restart?
5. Which evidence in the logs supports your conclusion?

Activity Steps

Step 1 – Human Analysis (10 minutes): Students manually review the logs and try to identify suspicious patterns.

Step 2 – AI Assistance (10 minutes): Students use an AI tool to analyze the logs and compare the results with their manual analysis.

Step 3 – Class Discussion: Discuss whether the incident was likely caused by a cyberattack, system overload, or software failure.